CSE 21 Fa26
Homework 4
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1. Let $(a_1, \ldots, a_n)$ be a random permutation of the elements $\{1, \ldots, n\}$ selected uniformly at random. Compute each expected value using linearity of expectation.

   (a) Compute the expected number of indices that have the property that $i < a_i$.
   Let X_i be the indicator that is 1 if $i < a_i$ and 0 otherwise, and let $X = \sum_{i=1}^{n} X_i$. The value a_i is equally likely to be any of the n elements, and the ones bigger than i are $i+1, \ldots, n$, so there are $n - i$ of them.

   $$P(X_i = 1) = \frac{n-i}{n}$$

   By linearity of expectation,

   $$E[X] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} \frac{n-i}{n} = \frac{1}{n} \sum_{i=1}^{n} (n-i)$$

   The sum $\sum_{i=1}^{n}(n-i)$ is $0 + 1 + \cdots + (n-1) = \dfrac{(n-1)n}{2}$, so

   $$E[X] = \frac{1}{n} \cdot \frac{(n-1)n}{2} = \frac{n-1}{2}$$

   (b) Compute the expected number of interior elements a_i such that $a_{i-1} < a_i < a_{i+1}$.
   An interior index is an index i with $2 \le i \le n-1$, so there are $n-2$ of them. Let X_i be 1 if $a_{i-1} < a_i < a_{i+1}$ and 0 otherwise. The three values a_{i-1}, a_i, a_{i+1} are three distinct numbers in a random permutation, so all $3! = 6$ orders are equally likely, and only one of those orders has $a_{i-1} < a_i < a_{i+1}$.

   $$P(X_i = 1) = \frac{1}{6}$$

   $$E[X] = \sum_{i=2}^{n-1} \frac{1}{6} = (n-2) \cdot \frac{1}{6} = \frac{n-2}{6}$$

   (c) Compute the expected number of swaps. A swap is a pair of elements (a_i, a_j) such that $a_i = j$ and $a_j = i$ and $i < j$.
   For each pair $i < j$, let X_{ij} be 1 if $a_i = j$ and $a_j = i$, and 0 otherwise. Position i is equally likely to be any of the n values, so $P(a_i = j) = \dfrac{1}{n}$. After that, position j is equally likely to be any of the remaining $n-1$ values, so $P(a_j = i \mid a_i = j) = \dfrac{1}{n-1}$.

   $$P(X_{ij} = 1) = \frac{1}{n} \cdot \frac{1}{n-1} = \frac{1}{n(n-1)}$$

   There are $\binom{n}{2}$ pairs with $i < j$, so

   $$E[X] = \binom{n}{2} \cdot \frac{1}{n(n-1)} = \frac{n(n-1)}{2} \cdot \frac{1}{n(n-1)} = \frac{1}{2}$$

2. (a) An art museum with 12 different rooms just got a shipment of 9 new different paintings. The museum curator puts the paintings in the rooms uniformly at random (each of the 12^9 arrangements is equally likely).

i. Compute the expected number of rooms without any paintings.
Let X_i be 1 if room i has no paintings and 0 otherwise. Each painting is placed independently, and it misses room i with probability $\frac{11}{12}$. All 9 paintings miss room i with probability

$$P(X_i = 1) = \left(\frac{11}{12}\right)^9$$

There are 12 rooms, so

$$E[X] = \sum_{i=1}^{12} \left(\frac{11}{12}\right)^9 = 12\left(\frac{11}{12}\right)^9 \approx 5.48$$

ii. Compute the expected number of rooms with exactly 1 painting.
Let X_i be 1 if room i has exactly one painting and 0 otherwise. We choose which 1 of the 9 paintings goes in room i, that painting lands in room i with probability $\frac{1}{12}$, and each of the other 8 paintings lands in one of the other 11 rooms.

$$P(X_i = 1) = 9 \cdot \frac{1}{12} \cdot \left(\frac{11}{12}\right)^8$$

$$E[X] = 12 \cdot 9 \cdot \frac{1}{12} \cdot \left(\frac{11}{12}\right)^8 = 9\left(\frac{11}{12}\right)^8 \approx 4.49$$

(b) An art museum with 12 different rooms just got a shipment of 9 indistinguishable prints. The museum curator puts the prints in the rooms uniformly at random (each of the $\binom{9+12-1}{12-1}$ arrangements is equally likely).
These arrangements are the non-negative integer solutions of $a_1 + a_2 + \cdots + a_{12} = 9$.
There are $\binom{20}{11}$ such arrangements.

i. Compute the expected number of rooms without any prints.
Let X_i be 1 if room i has no prints and 0 otherwise. If room i is empty, the 9 prints are placed in the other 11 rooms. The number of ways to do that is

$$\binom{9+11-1}{11-1} = \binom{19}{10}$$

$$P(X_i = 1) = \frac{\binom{19}{10}}{\binom{20}{11}}$$

$$\frac{\binom{19}{10}}{\binom{20}{11}} = \frac{19!}{10!\,9!} \cdot \frac{11!\,9!}{20!} = \frac{11}{20}$$

$$E[X] = \sum_{i=1}^{12} \frac{11}{20} = 12 \cdot \frac{11}{20} = \frac{33}{5} = 6.6$$

ii. Compute the expected number of rooms with exactly 1 print.
Let X_i be 1 if room i has exactly one print and 0 otherwise. If room i has exactly one print, the remaining 8 prints are placed in the other 11 rooms. The number of ways to do that is

$$\binom{8+11-1}{11-1} = \binom{18}{10}$$

$$P(X_i = 1) = \frac{\binom{18}{10}}{\binom{20}{11}}$$

$$\frac{\binom{18}{10}}{\binom{20}{11}} = \frac{18!}{10!\,8!} \cdot \frac{11!\,9!}{20!} = \frac{11 \cdot 9}{20 \cdot 19} = \frac{99}{380}$$

$$E[X] = 12 \cdot \frac{99}{380} = \frac{297}{95}$$

3. (a) What is the expected number of coin flips of a fair coin until you get two consecutive H's?
We first divide the possible starting outcomes into 3 cases:

$$P(T) = \frac{1}{2}, \qquad P(HT) = \frac{1}{4}, \qquad P(HH) = \frac{1}{4}.$$

Now we find each conditional expected value:

- $E[X \mid T] = 1 + E[X]$: we used 1 flip but have to start over.
- $E[X \mid HT] = 2 + E[X]$: used 2 flips but have to start over.
- $E[X \mid HH] = 2$: used 2 flips and the goal was achieved.

Using

$$E[X] = E[X \mid T]P(T) + E[X \mid HT]P(HT) + E[X \mid HH]P(HH)$$

gives

$$E[X] = \frac{1}{2}(1 + E[X]) + \frac{1}{4}(2 + E[X]) + \frac{1}{4}(2).$$

$$E[X] = \frac{3}{2} + \frac{3}{4}E[X]$$

$$\frac{1}{4}E[X] = \frac{3}{2}$$

$$E[X] = 6$$

(b) What is the expected number of coin flips until you get HHH?
First find the probability of the 4 cases:

$$P(T) = \frac{1}{2}, \qquad P(HT) = \frac{1}{4}, \qquad P(HHT) = \frac{1}{8}, \qquad P(HHH) = \frac{1}{8}.$$

Now find each conditional expected value:

- $E[X \mid T] = 1 + E[X]$: 1 flip, start over.
- $E[X \mid HT] = 2 + E[X]$: 2 flips and start over.
- $E[X \mid HHT] = 3 + E[X]$: 3 flips, start over.
- $E[X \mid HHH] = 3$: goal achieved after 3 flips.

$$E[X] = \frac{1}{2}(1 + E[X]) + \frac{1}{4}(2 + E[X]) + \frac{1}{8}(3 + E[X]) + \frac{1}{8}(3)$$

$$E[X] = \frac{7}{4} + \frac{7}{8}E[X]$$

$$\frac{1}{8}E[X] = \frac{7}{4}$$

$$E[X] = 14$$

4. An ideal hash function from $\{1, \ldots, m\}$ to $\{1, \ldots, n\}$ is a function selected uniformly at random from the set of all functions with that domain and codomain.

(a) Let H be an ideal hash function from $\{1, \ldots, m\}$ to $\{1, \ldots, n\}$. What is the expected number of collisions? A collision is a pair of domain elements $i < j$ such that $H(i) = H(j)$.

$$X_{ij} = \begin{cases} 1 & \text{if } H(i) = H(j) \\ 0 & \text{otherwise} \end{cases}$$

Let $X = \sum_{1 \le i < j \le m} X_{ij}$, so

$$E[X] = \sum_{1 \le i < j \le m} E[X_{ij}] = \sum_{1 \le i < j \le m} P(H(i) = H(j)).$$

For m distinct inputs we have $\binom{m}{2}$ pairs.

$$P(H(i) = H(j)) = \frac{n}{n^2} = \frac{1}{n}$$

$$E[X] = \binom{m}{2}\frac{1}{n}$$

(b) Let H be an ideal hash function from $\{1, \ldots, m\}$ to $\{1, \ldots, n\}$. What is the expected number of codomain values that do not get mapped to?

$$X_j = \begin{cases} 1 & \text{if no input hashes to } j \\ 0 & \text{otherwise} \end{cases}$$

Since there are n possible hash values,

$$E[X] = n\,P(X_1 = 1).$$

For each particular hash value j, each input avoids it with probability

$$P(H(i) \neq j) = 1 - \frac{1}{n}.$$

The probability that all m inputs avoid j is

$$P(X_j = 1) = \left(1 - \frac{1}{n}\right)^m.$$

$$E[X] = n\left(1 - \frac{1}{n}\right)^m$$